Newman-Penrose Formalism Calculations

Stephani et al., Equation (18.66)

Metric

$$ds^2 = -\frac{x^2}{N(z)^2}\mathrm{d}t \otimes \mathrm{d}t + \frac{1}{(ax^2 + b\log(x) + c)N(z)^2}\mathrm{d}x \otimes \mathrm{d}x + \frac{1}{N(z)^2}\mathrm{d}z \otimes \mathrm{d}z + \left(\frac{ax^2 + b\log(x) + c}{N(z)^2}\right)\mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, x, z, \phi)$$

Metric Functions

$$G = ax^2 + b\log(x) + c$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}x}{2N(z)}\right) dt + \left(\frac{\sqrt{2}}{2\sqrt{ax^2 + b\log(x) + c}N(z)}\right) dx$$

$$n = \left(\frac{\sqrt{2}x}{2N(z)}\right) dt + \left(-\frac{\sqrt{2}}{2\sqrt{ax^2 + b\log(x) + c}N(z)}\right) dx$$

$$m = \left(\frac{\sqrt{2}}{2N(z)}\right) dz + \left(\frac{i\sqrt{2}\sqrt{ax^2 + b\log(x) + c}}{2N(z)}\right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2N(z)}\right) dz + \left(-\frac{i\sqrt{2}\sqrt{ax^2 + b\log(x) + c}}{2N(z)}\right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}N\left(z\right)}{2\,x}\right)\,dt + \left(\frac{1}{2}\,\sqrt{2}\sqrt{ax^{2} + b\log\left(x\right) + c}N\left(z\right)\right)\,dx$$

$$n = \left(-\frac{\sqrt{2}N\left(z\right)}{2\,x}\right)\,dt + \left(-\frac{1}{2}\,\sqrt{2}\sqrt{ax^{2} + b\log\left(x\right) + c}N\left(z\right)\right)\,dx$$

$$m = \left(\frac{1}{2}\,\sqrt{2}N\left(z\right)\right)\,dz + \left(\frac{i\,\sqrt{2}N\left(z\right)}{2\,\sqrt{ax^{2} + b\log\left(x\right) + c}}\right)\,d\phi$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2}N\left(z\right)\right)\,dz + \left(-\frac{i\,\sqrt{2}N\left(z\right)}{2\,\sqrt{ax^{2} + b\log\left(x\right) + c}}\right)\,d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}N\left(z\right)}{2\,x}\,\partial_t + \frac{1}{2}\,\sqrt{2}\sqrt{ax^{2} + b\log\left(x\right) + c}N\left(z\right)\,\partial_x$$

$$\Delta = -\frac{\sqrt{2}N\left(z\right)}{2\,x}\,\partial_t - \frac{1}{2}\,\sqrt{2}\sqrt{ax^{2} + b\log\left(x\right) + c}N\left(z\right)\,\partial_x$$

$$\delta = \frac{1}{2}\,\sqrt{2}N\left(z\right)\,\partial_z + \frac{i\,\sqrt{2}N\left(z\right)}{2\,\sqrt{ax^{2} + b\log\left(x\right) + c}}\,\partial_\phi$$

$$\overline{\delta} = \frac{1}{2}\,\sqrt{2}N\left(z\right)\,\partial_z - \frac{i\,\sqrt{2}N\left(z\right)}{2\,\sqrt{ax^{2} + b\log\left(x\right) + c}}\,\partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}axN\left(z\right)}{4\sqrt{ax^{2} + b\log\left(x\right) + c}} - \frac{\sqrt{2}bN\left(z\right)}{8\sqrt{ax^{2} + b\log\left(x\right) + c}x}$$

$$\sigma = \frac{\sqrt{2}axN\left(z\right)}{4\sqrt{ax^{2} + b\log\left(x\right) + c}} + \frac{\sqrt{2}bN\left(z\right)}{8\sqrt{ax^{2} + b\log\left(x\right) + c}x}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}axN\left(z\right)}{4\sqrt{ax^{2} + b\log\left(x\right) + c}} - \frac{\sqrt{2}bN\left(z\right)}{8\sqrt{ax^{2} + b\log\left(x\right) + c}x}$$

$$\lambda = \frac{\sqrt{2}axN\left(z\right)}{4\sqrt{ax^{2} + b\log\left(x\right) + c}} + \frac{\sqrt{2}bN\left(z\right)}{8\sqrt{ax^{2} + b\log\left(x\right) + c}x}$$

$$\nu = 0$$

$$\alpha = \frac{1}{4}\sqrt{2}\frac{\partial}{\partial z}N\left(z\right)$$

$$\beta = -\frac{1}{4}\sqrt{2}\frac{\partial}{\partial z}N\left(z\right)$$

$$\tau = \frac{1}{2}\sqrt{2}\frac{\partial}{\partial z}N\left(z\right)$$

$$\pi = -\frac{1}{2}\sqrt{2}\frac{\partial}{\partial z}N\left(z\right)$$

$$\epsilon = \frac{\sqrt{2}\sqrt{ax^{2} + b\log\left(x\right) + c}N\left(z\right)}{4x}$$

$$\gamma = \frac{\sqrt{2}\sqrt{ax^{2} + b\log\left(x\right) + c}N\left(z\right)}{4\,x}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{bN\left(z\right)^{2}}{4\,x^{2}}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = \frac{1}{2}\,aN\left(z\right)^{2} + \frac{1}{2}\,N\left(z\right)\frac{\partial^{2}}{(\partial z)^{2}}N\left(z\right)$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{4}\,aN\left(z\right)^{2} + \frac{1}{4}\,N\left(z\right)\frac{\partial^{2}}{(\partial z)^{2}}N\left(z\right) + \frac{bN\left(z\right)^{2}}{8\,x^{2}}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = \frac{1}{2}\,aN\left(z\right)^{2} + \frac{1}{2}\,N\left(z\right)\frac{\partial^{2}}{(\partial z)^{2}}N\left(z\right)$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{bN\left(z\right)^{2}}{4\,x^{2}}$$

$$\Lambda = -\frac{1}{4}\,aN\left(z\right)^{2} - \frac{1}{2}\,\frac{\partial}{\partial z}N\left(z\right)^{2} + \frac{1}{4}\,N\left(z\right)\frac{\partial^{2}}{(\partial z)^{2}}N\left(z\right) - \frac{bN\left(z\right)^{2}}{24\,x^{2}}$$

Weyl Tensor Components

$$\Psi_0 = -\frac{bN\left(z\right)^2}{4\,x^2}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{bN\left(z\right)^2}{12\,x^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = -\frac{bN\left(z\right)^2}{4\,x^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"